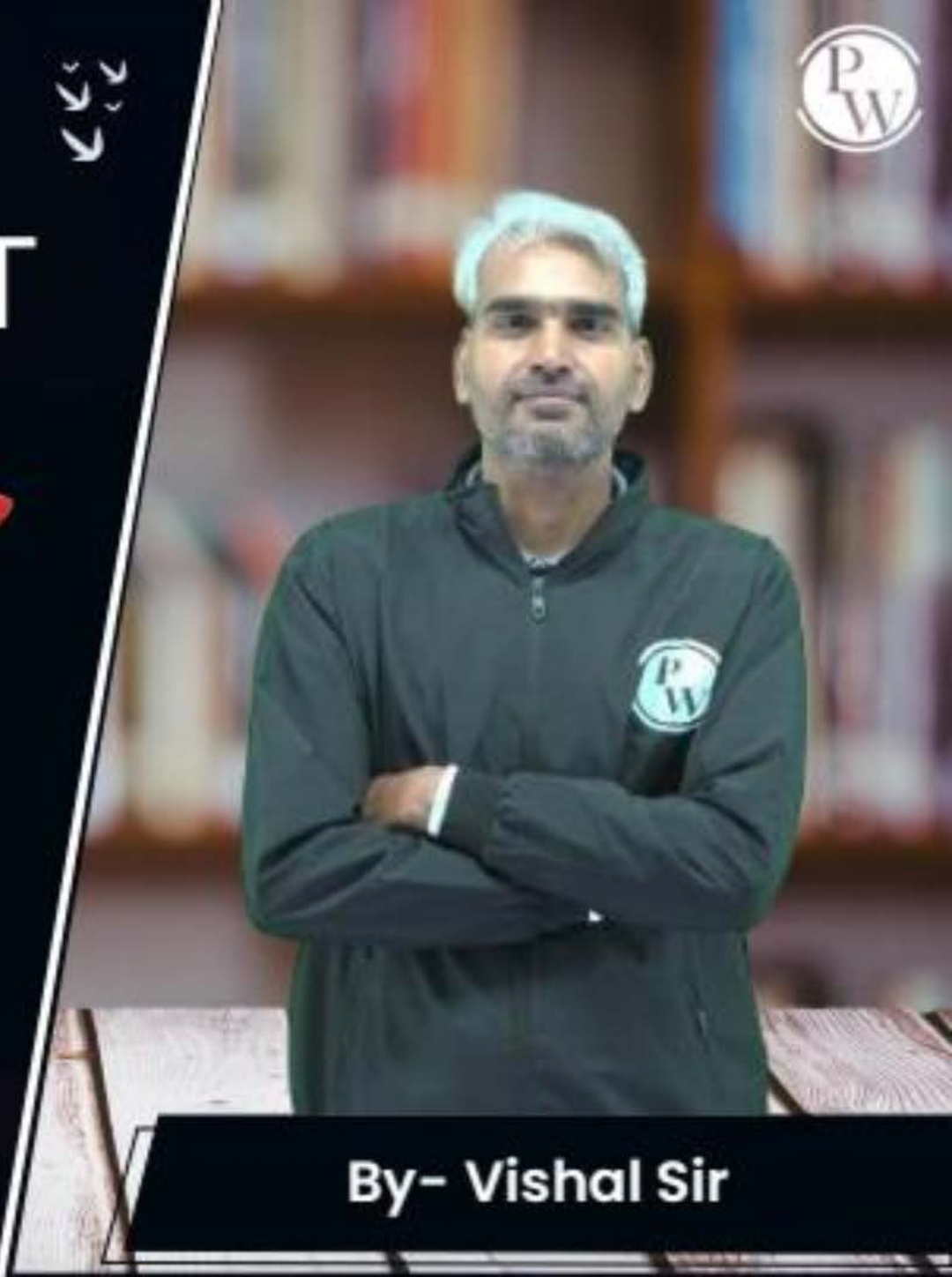
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture



Topic Properties of functional dependency

Topic Different types of keys in RDBMS

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Topics to be Covered



- Topic — Candidate key
- Topic — Primary key
- Topic — Alternate key
- Topic — Super key
- Topic — Closure of an attribute set

Slide



Topic : Properties of Functional Dependencies

① Reflexivity :→ If $X \supseteq Y$, then $X \longrightarrow Y$ is called a reflexive FD.

Every Reflexive FD always holds in a relation

② Augmentation :→ Let functional dependency $X \longrightarrow Y$ exists in the relation R, and Z is some set of attributes from relation R, then

$XZ \longrightarrow YZ$ will also exist in rel^n R.

③ Transitivity :→ Let FDs $X \longrightarrow Y$ & $Y \longrightarrow Z$ exists in the relation, R, then FD $X \longrightarrow Z$ will also exist in the relation R.

This three properties are called Armstrong's Axioms



Topic : Properties of Functional Dependencies

④ Decomposition :- (Splitting Rule)

If $X \rightarrow YZ$ exists in relation R, then $X \rightarrow Y$ and $X \rightarrow Z$ both will hold true in relation R.

Note:- If $XY \rightarrow Z$ exists in R,

then $X \rightarrow Z$ & $Y \rightarrow Z$ need not hold true in rel^n R

Let $AB \rightarrow BC$ holds true in rel^n R

Semi non-trivial FD

$AB \rightarrow BC \implies$ $AB \rightarrow B$ {trivial Part}

&

$AB \rightarrow C$ {non-trivial}

We can always split a semi non-trivial FD into two Parts, trivial Part & non-trivial Part.



Topic : Properties of Functional Dependencies

⑤ Union :- If $X \rightarrow Y$ and $X \rightarrow Z$ exists in relation R, then $X \rightarrow YZ$ will also hold true in relation R

⑥ Composition :- If $X \rightarrow Y$ and $P \rightarrow Q$ exists in relation R then $XP \rightarrow YQ$ will also hold in relation R

⑦ Pseudo Transitivity :- If $X \rightarrow Y$ and $YW \rightarrow Z$ exists in relation R then $XW \rightarrow Z$ will also hold true in rel^n R

Key Concept :-

In a relational table no two tuples of the table can be exactly same

↳ i.e, in relational table duplicate tuples are not allowed.

↳ To implement this restriction we say that every relation must have a key.

* A key in a relation is the set of attributes that can uniquely identify each tuple of the relation



Topic : Different types of keys

There are various types of keys

1. Candidate key
2. Primary key
3. Alternate key
4. Super key
5. Foreign key

Foreign key Concept will be discussed in the last Capter
i.e: "ER Model & Integrity Constraints"

* **Minimal Set** :→ A set of elements from which no element can be removed without loosing the associated property is called a minimal set.

i.e. Whenever we remove any element from the set then it is guaranteed that it will lose its property

Note :→ If values of a set of attributes are guaranteed to be unique in a relation, then that set of attributes is definitely a "key" of that relation {But it may or may not be minimal}



Topic : Candidate key

i.e. no attribute can be removed from the set without destroying its property of being a key {i.e. without destroying its property of being unique}

A minimal set of attributes that can identify each tuple of the relation uniquely is called a Candidate key

(or)

A minimal set of attributes that can determine each attribute of the relation is called a Candidate key.

eg①:- Consider the following relation

Student

Sid	Sname	Fee
S_1	A	500
S_2	A	500
S_3	B	600
S_4	B	400
S_5	C	600

Also consider that following functional dependencies holds in the relation

$\{ Sid \rightarrow Sname \}$
$Sid \rightarrow fee \}$

$Sid \rightarrow Sname, fee, Sid$

* With respect to given FDs we can observe that values of Sid will always be unique in the relation Student Hence, "Sid" will be a key of the relation "Student"
* If values of Sid are unique, then values of (Sid, Sname) together will also be unique Hence, (Sid, Sname) is also a key of "Student"

eg②: Consider the following relation

Enroll

Sid	Cid	I-id
S_1	C_1	101
S_2	C_1	101
S_3	C_2	101
S_3	C_3	102
S_4	C_3	102

Consider following functional dependency holds in the relation

$\{ Cid \rightarrow I\text{-}id \}$
$\Downarrow$
$Sid, Cid \rightarrow Sid, I\text{-}id$
$\Downarrow$
$Sid, Cid \Rightarrow Sid, Cid, I\text{-}id$

* Together the values of (Sid, Cid) will be unique in the relation, Hence (Sid, Cid) is a key of the relation Enroll.
* Values of (Sid, Cid) are unique, ∴ Values of (Sid, Cid, I-id) will also be unique. Hence, (Sid, Cid, I-id) is also a key of relation "Enroll"

eg①:- Consider the following relation

Student

Sid	Sname	Fee
S_1	A	500
S_2	A	500
S_3	B	600
S_4	B	400
S_5	C	600

Also consider that following functional dependencies holds in the relation

{ Sid → Sname }
{ Sid → fee }

Sid → Sname, fee, Sid

* With respect to given FDs we can observe that values of Sid will always be unique in the relation Student Hence, "Sid" will be a key of the relation "Student"
* If values of Sid are unique, then values of (Sid, Sname) together will also be unique Hence, (Sid, Sname) is also a key of "Student"

In eg①:-

(Sid, Sname) is a key of the relation Student, but it is not minimal, Because if we remove "Sname" from the set then "Sid" alone can uniquely identify each tuple of the relation

∴ (Sid, Sname) is not a candidate key of the relation

In eg① :- (Sid) is also a key of the relation, and it is minimal as well,

∴ (Sid) is a candidate key of relation Student

eg②: Consider the following relation

Enroll

Sid	Cid	I-id
S_1	C_1	101
S_2	C_1	101
S_3	C_2	101
S_3	C_3	102
S_4	C_3	102

Consider following functional dependencies holds in the relation

$\{ Cid \longrightarrow I\text{-}id \}$

$\Downarrow$

$Sid, Cid \longrightarrow Sid, I\text{-}id$

$\Downarrow$

$Sid, Cid \Rightarrow Sid, Cid, I\text{-}id$

- Together the values of (Sid, Cid) will be unique in the relation,
  Hence (Sid, Cid) is a key of the relation Enroll.
- Values of (Sid, Cid) are unique, ∴ Values of (Sid, Cid, I-id) will also be unique.
  Hence, (Sid, Cid, I-id) is also a key of relation "Enroll"

In eg②:

(Sid, Cid) is a key of the relation Enroll.

- If we remove Sid from the set, then "Cid" alone can not uniquely identify each tuple of relation.

and if remove "Cid" from the set, then "Sid" alone can not uniquely identify each tuple of the relation.

i.e, no attribute can be removed from the set without destroying its property of being a key

∴ (Sid, Cid) is a minimal key and Hence a Candidate key of relation "Enroll".

- In eg ②:- (Sid, Cid, I-id) is also a key of relation, but it is not minimal because "I-id" can be removed from the set without destroying its property of being a key.

∴ (Sid, Cid, I-id) is not a Candidate key.



Topic : Candidate key

* If a key is formed of a single attribute then it is always minimal, and hence a candidate key

① If Candidate key is formed of a single attribute. then such candidate keys are called simple Candidate key

② If Candidate key is formed of two or more attributes then such candidate keys are called Compound or Composite Candidate key.

③ A relation may have more than one candidate key.
{The number of candidate keys in a relation will be based on the functional dependencies that holds in that relation}



Topic : Candidate key

④ Attributes belonging to any of the Candidate key of the relation are called "prime attributes" of the relation.

⑤ Attributes that does not belong to any of the Candidate key are called Non-prime attributes.



Topic : Primary key

- There may be multiple candidate keys in a relation. And one of those candidate key may be defined as "primary key" of that relation
- In a relation we can have at most one primary key.
- Attributes of primary key are not allowed to take "NULL" values.



Topic : Alternate key

{ Also known a Secondary key }

All candidate keys Except primary key are called atlernate keys



Topic : Super key

"need not be minimal"

A set of attributes that uniquely identify each tuple of the relation is called a super key.

(or)

A set of attributes (need not be minima) that can determine all attributes of the relation is called a Super key.

* Any super-set of a candidate key is a Super key.
* Every candidate key is also a Super key, but Every Super key need not be a candidate key.

Note:- At least one subset of every Super key is guaranteed to be a candidate key.

Note:-

$$\text{Super key} = (\text{Attributes of any one of the candidate key}) + (\text{0 or more attributes out of remaining attributes})$$

eg: Consider the following relational schema

Student (Sid, Sname, Branch)

If "Sid" is the only candidate key of the relation Student, then identify the total number of superkeys in relation Student.

* Any superset of {Sid} will be a Superkey

Supersets of {Sid} =

- {Sid} } C.K + 0 out of remaining two
- {Sid, Sname} } C.K + 1 attribute out of remaining two
- {Sid, Branch} } C.K + 1 attribute out of remaining two
- {Sid, Sname, Branch} } C.K + 2 out of remaining two

← This are the 4 superkeys of relation Student.



Topic : Closure of an attribute set

Closure of an attribute set X (i.e., X^+) can be defined as set of all the attributes that can be functionally determined from attributes of set X.

#e.g. Assume a relation R (A,B,C,D) that has the following functional dependencies:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$



Topic : Super key

Let R be the relational schema, and let X be some set of attributes over relation R. If X^+ determines all attributes of relation R, then X is called super key of relation R.

#e.g. Assume a relation R (A,B,C,D) that has the following functional dependencies:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$



Topic : Candidate key (Minimal Super key)

Let R be the relational schema, and let X be the super key of relation R.

If no proper subset of X is a super key the X is minimal super key i.e., X is Candidate key

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

AB→C,

B→E,

C→D

Find the Candidate key of R.

#Q. Assume a relation R (A, B, C, D, E) that has the following functional dependencies:

AB→C,

B→E,

C→D

E→A

Find the Candidate key of R.



2 mins Summary

- Topic: Candidate key
- Topic: Primary key
- Topic: Alternate key
- Topic: Super key
- Topic: Closure of an attribute set

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THANK - YOU

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